

Data Preprocessing & Analysis Guide

1. Understanding Data Analysis

Data analysis involves examining raw information to discover patterns, relationships, and useful insights. In this context, it helps evaluate customer financial behavior and identify potential default risks.

2. Data Collection

Data can originate from multiple sources such as structured files, databases, and APIs. Combining these sources enhances the completeness and reliability of the dataset.

3. Cleaning the Dataset

Before modeling, the dataset must be cleaned by removing duplicates, correcting inconsistencies, and ensuring proper data formats. Clean data ensures better results.

4. Handling Missing Data

Missing values can be treated using techniques like mean/median replacement, most frequent values, predictive methods like KNN, or removing incomplete rows when necessary.

5. Managing Outliers

Extreme values can distort analysis. Methods like IQR, Z-score, and capping techniques help control their impact without losing valuable data.

6. Feature Creation

New meaningful variables can be derived from existing data such as ratios or aggregated metrics. These improve model understanding and prediction accuracy.

7. Variable Types

Data may include numerical, categorical, and date-based features. Date variables can be broken into components like year or month for better analysis.

8. Encoding Categories

Categorical data must be converted into numeric form using methods like one-hot encoding, label encoding, or ordinal encoding.

9. Transforming Numerical Features

Numerical values can be grouped or converted using binning or thresholding to simplify patterns and improve interpretability.

10. Scaling Features

Scaling ensures all numerical features are on a similar range using techniques like standardization or normalization, which improves model performance.

11. Data Transformation Techniques

Transformations such as log or square root help normalize skewed data and stabilize variance for better modeling.

12. Column-wise Processing

Different preprocessing steps can be applied to different columns using structured pipelines, making workflows efficient and organized.

13. Final Dataset

After preprocessing, the dataset becomes clean, structured, and ready for machine learning tasks.

Conclusion

A proper preprocessing pipeline ensures data quality, reduces noise, and enhances the performance of machine learning models in real-world applications.